

# DOT - How It Works and How to Use It

A concise guide to the Dipole Optimization Tool

## Purpose

DOT automatically synthesizes two-dimensional superconducting dipole cross-sections from electromagnetic targets, cable data, and geometric constraints. It searches for a Pareto set instead of copying a reference layout.

## DOT calculates

Block geometry, target-field current, bore harmonics, peak field, short-sample current, load-line margin, and geometric feasibility.

## DOT does not calculate

Iron, coil ends, persistent-current effects, mechanics, protection, tolerances, cryogenics, or any three-dimensional behavior.

**Correct interpretation: a certified candidate satisfies the declared DOT equations and straight-section constraints during final verification. It is an electromagnetic pre-design, not a construction release.**

**Author:** Mattia Elisei

INFN Milan and University of Rome La Sapienza

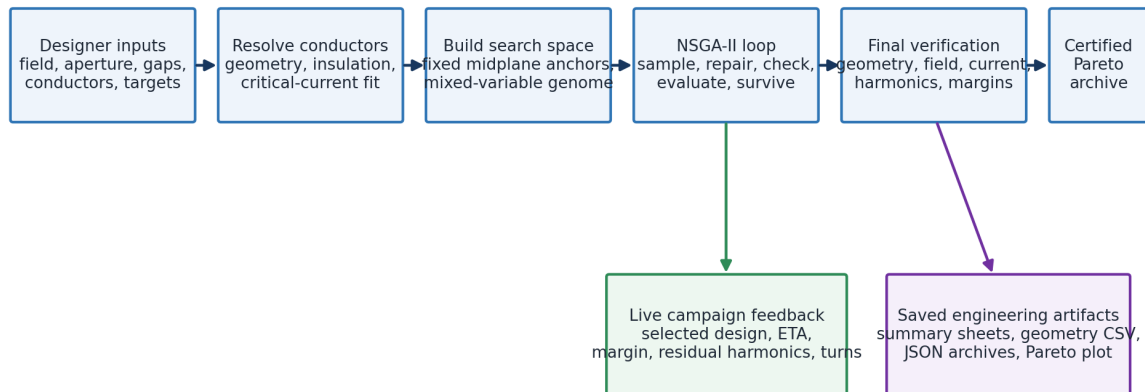
**Document date:** 29 July 2026

## PART I - HOW IT WORKS

# One workflow, from requirements to a certified Pareto archive

The optimizer never receives a known coil layout. It receives only the design space, the conductor data, and the required performance.

## DOT end-to-end workflow



The optimizer never receives a reference coil layout. It receives only the declared design space and requirements.

| Stage      | What happens                                                                                                      | Why it matters                                                  |
|------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 1. Define  | Field, aperture, cables, layers, gaps, block/turn bounds, current and acceptance targets.                         | These inputs determine which designs can exist.                 |
| 2. Resolve | Each conductor name is linked through the cadata catalogue to cable, strand, insulation and critical-current fit. | Geometry and load-line physics use one consistent data chain.   |
| 3. Search  | NSGA-II breeds, repairs, checks and evaluates mixed continuous, integer and binary genomes.                       | Field quality and margin are explored together without weights. |
| 4. Certify | The final hard-feasible archive is recalculated with fixed final settings and every target is re-applied.         | Only this stage can label a candidate certified.                |
| 5. Export  | DOT saves the Pareto front, representative design, alternative topologies, geometry tables and provenance.        | The result can be inspected and reproduced.                     |

### The essential distinction

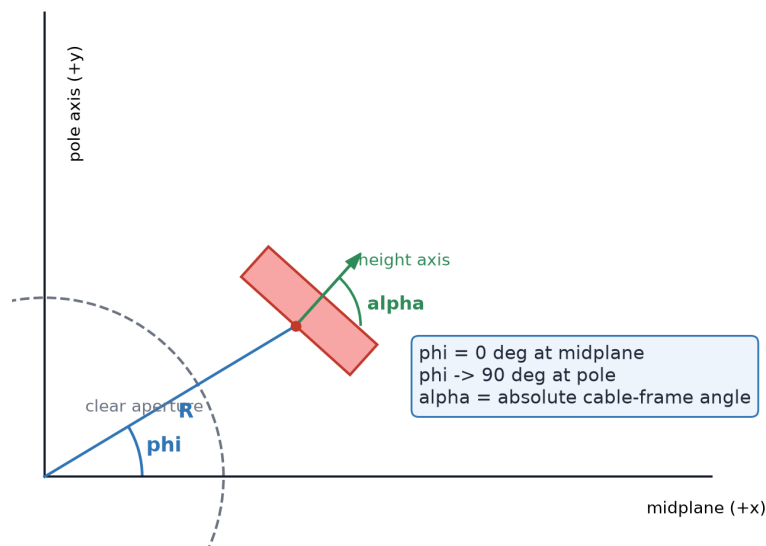
Search values guide evolution. Final checks decide acceptance. A live best candidate is useful progress information, but it is not a final pass until the end-of-run checks succeed.

## PART I - HOW IT WORKS

# Geometry: fixed anchors, variable blocks

DOT works in the first quadrant. The midplane is +x, the pole direction is +y, phi starts at zero on the midplane, and alpha is the absolute cable-frame angle.

## Quadrant geometry and angle convention



### Layer 1 anchor

**R1 = aperture radius**

$\phi_1 = \text{atan}(\text{azimuthal gap} / R_1)$

$\alpha_1 = 0 \text{ deg}$

### Later-layer anchor

R is the outermost x coordinate of the previous layer's first insulated turn plus the radial gap. Then  $\phi = \text{atan}(\text{gap} / R)$ ,  $\alpha = 0 \text{ deg}$ .

| Quantity       | Midplane block in every layer                | All later block slots                      |
|----------------|----------------------------------------------|--------------------------------------------|
| Existence      | Always active                                | Optimizer may activate a contiguous prefix |
| Layer radius R | Derived from aperture and radial gaps; fixed | Same fixed layer radius                    |
| Turns          | Integer chosen by optimizer                  | Integer chosen when active                 |
| phi            | Derived from azimuthal gap; fixed            | Continuous optimizer variable              |
| alpha          | Fixed at 0 deg                               | Continuous optimizer variable              |

**Real geometry:** every insulated turn is a convex quadrilateral built from the cable dimensions, keystone, insulation, R, phi and alpha. Turns are arc-stacked from midplane toward pole. Block numbers run continuously across layers, beginning with Layer 1's midplane block.

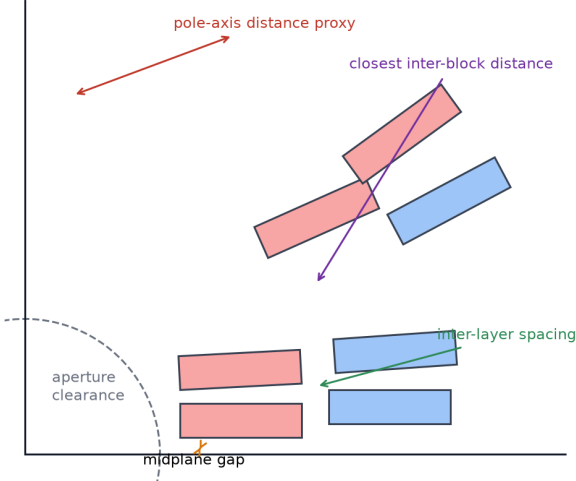
PART I - HOW IT WORKS

# Feasibility is decided on the actual insulated turn polygons

Constructive placement and repair make the search efficient; exact geometric checks remain the ground truth in every generation and again during certification.

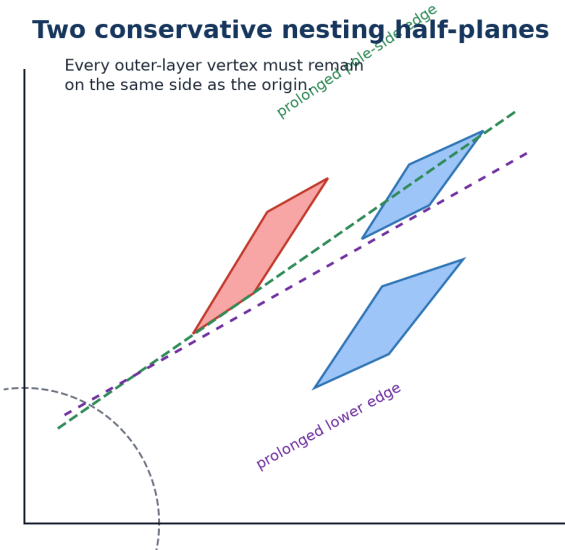
### Clearances and polygon checks

All checks use insulated convex turn polygons.



### Two conservative nesting half-planes

Every outer-layer vertex must remain on the same side as the origin.



| Rule                  | What DOT verifies                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| Aperture and midplane | No insulated polygon enters the clear bore or crosses the symmetry plane.                                         |
| Turn collision        | No positive-area intersection within a block, between blocks, or between layers.                                  |
| Inter-block clearance | Exact closest polygon distance in each layer is at least the requested value.                                     |
| Radial separation     | Adjacent layer radii preserve the requested insulated radial build.                                               |
| Pole-turn proxy       | The closest Layer-1 insulated point stays at least the requested distance from the pole axis.                     |
| Layer nesting         | Every outer-layer vertex remains on the aperture side of both prolonged long-edge lines from the inner pole turn. |
| Topology and turns    | Active blocks are contiguous and all block/layer/total turn limits are respected.                                 |

### Nesting in plain language

Imagine extending both long sides of the inner layer's pole-most turn as infinite lines. The complete outer layer must remain behind both lines, toward the aperture. This conservative envelope rejects layouts that may fit on paper but are unfavorable to wind as nested layers.

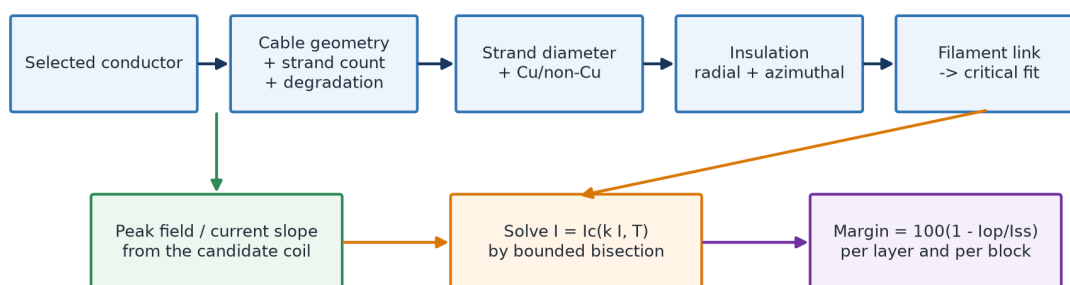
**Numerical tolerance is not a design gap.** It only classifies round-off and near-contact. Reserve the physical clearance you actually need.

## PART I - HOW IT WORKS

# Physics: geometry -> field -> harmonics -> load-line margin

The forward engine assumes infinitely long z-directed currents, exact dipole mirror symmetry, linear materials and no iron.

## Conductor catalogue resolution and load-line calculation



| Calculation          | Meaning                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bore field           | Biot-Savart fields from distributed line-current sources are summed at the center. In range mode, linearity gives the exact current required for the target field. |
| Harmonics            | Normal coefficients $b_3, b_5, \dots$ are reported in accelerator units at the reference radius: one unit is $1e-4$ of the main normal dipole field.               |
| Peak field           | DOT samples the bare conductor boundary. The most exposed block/layer controls its load line.                                                                      |
| Short-sample current | For each conductor, DOT intersects the geometry's peak-field/current slope with the linked critical-current surface at the selected temperature.                   |
| Margin               | $\text{margin} = 100 \times (1 - I_{op} / I_{ss})$ . DOT reports it per block and layer and optimizes the minimum value.                                           |

### Current range mode

If Min Current and Max Current differ, DOT scales the series current to the requested bore field, then enforces both bounds.

### Fixed-current mode

If Min Current equals Max Current, DOT applies that exact current and requires the geometry to reproduce the bore field within 0.01%.

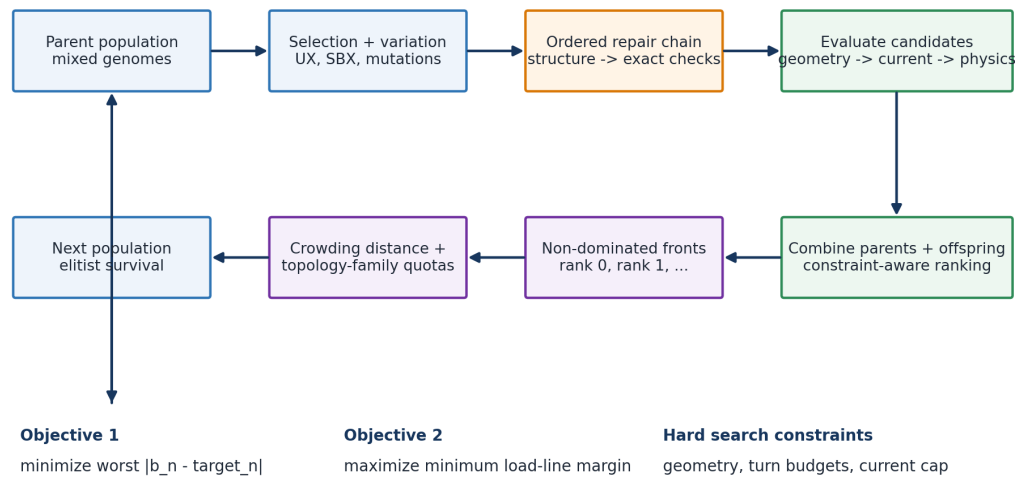
Supported cadata critical surfaces are REMFIT type 1 (Nb-Ti) and type 11 (Nb3Sn). The conductor name must exactly match the selected catalogue record. The GUI's current advice is a conservative Layer-1 sanity check, not a hard limit.

## PART I - HOW IT WORKS

# NSGA-II keeps the harmonic-margin trade-off visible

DOT has two electromagnetic objectives and no hidden weighted sum: minimize the worst requested harmonic residual and maximize the minimum load-line margin.

## DOT's constrained, topology-aware NSGA-II generation



Final acceptance thresholds are re-applied during final verification; the live best is one selected view of the population.

| Concept               | Meaning in DOT                                                                                                                            |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Dominance             | A candidate dominates another only if it is no worse in both objectives and better in at least one.                                       |
| Hard constraints      | Geometry, turn budgets and current feasibility take priority over objective quality.                                                      |
| Diversity             | Crowding distance spreads survivors along the front; topology-family quotas keep distinct block-count patterns alive.                     |
| Variation             | Parent selection, crossover and mutation change active blocks, turns, phi and alpha; repair restores ordered, packable winding structure. |
| Complexity preference | Fewer turns, then fewer blocks, are preferred only when electromagnetic objectives are numerically equivalent.                            |

### Final certification

After the last generation DOT deduplicates the hard-feasible archive and recalculates it with fixed final settings (12 x 12 bore and 80 x 80 peak-field midpoint sources per turn), re-applies every target, and rebuilds the non-dominated certified front.

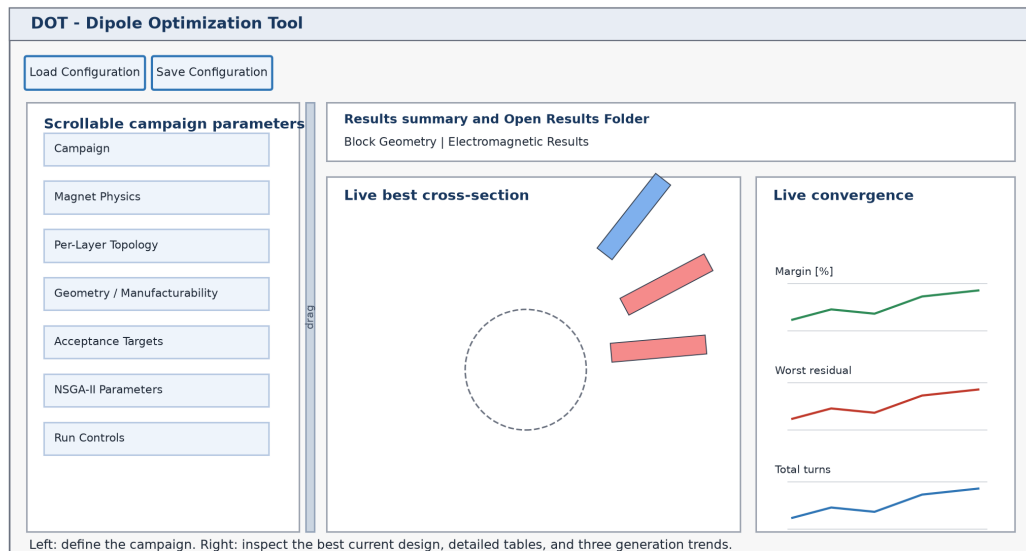
### Optional radial-block preference (1.1.0)

**Off by default.** After target-met candidates persist for three generations, DOT uses 5% of later offspring to move only free alpha angles toward central-cable radial alignment. The complete repair and geometry checks run again. At most one radial exemplar from the best Pareto rank is retained; bore field, harmonics, load-line margin, powering limits, and geometry keep precedence.

## PART II - HOW TO USE IT

# Launch once, then configure the campaign in a physical order

On Windows, double-click `launch_dot_gui.cmd`. Python is the only prerequisite; the launcher lists dependencies, asks permission, creates `.venv`, installs them, and opens DOT.



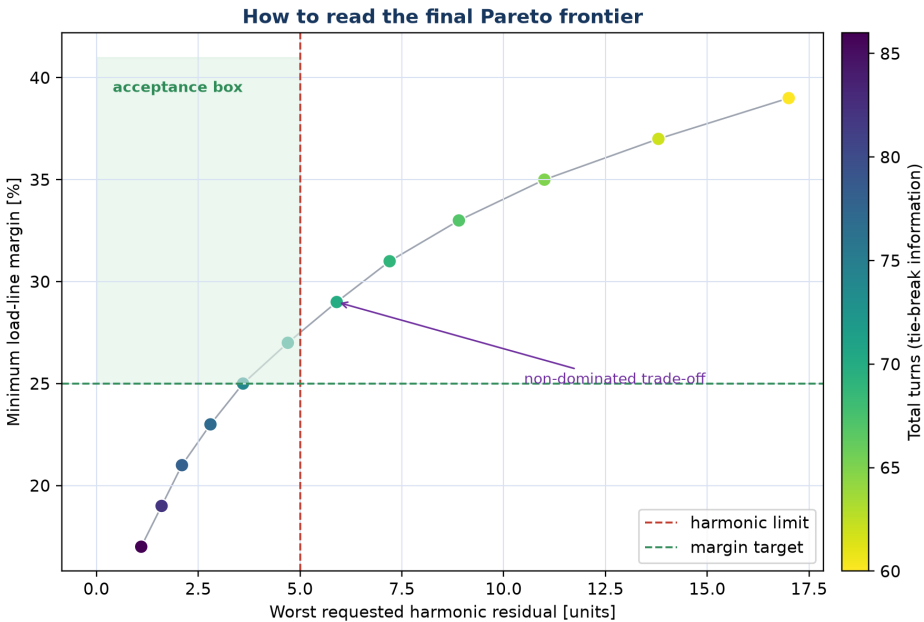
| Order | Set                                                         | Good practice                                                                                                                |
|-------|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 1     | Campaign name and output directory                          | Use a new descriptive name; every run receives a timestamped folder.                                                         |
| 2     | Field, aperture, temperature, reference radius, layers      | Start with $R_{ref} = \text{round}(2/3 \text{ aperture radius}, 3)$ ; change it only for a defined field-quality convention. |
| 3     | One cadata file and supported conductor per layer           | Use the selector; the conductor name must match the file exactly.                                                            |
| 4     | Min/Max blocks and Min/Max Turns                            | Keep enough independent blocks for harmonic control without forcing unnecessary topology.                                    |
| 5     | Azimuthal/radial gaps, block clearance, pole proxy, nesting | Enter physical manufacturing clearances; Layer 1 has no radial-gap input.                                                    |
| 6     | Harmonic targets, margin and current bounds                 | Signed harmonic targets can pre-compensate a separately calculated yoke contribution.                                        |
| 7     | Search effort, seed and optional parallel evaluation        | Use a fixed seed for reproducibility; test parallel mode on the actual hardware.                                             |

|                                                                                                     |                                                                           |                                                                                             |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Quick exploration</b><br>160 population x 80 generations. Check reachability and input mistakes. | <b>Design search</b><br>480 x 200. Recommended normal synthesis campaign. | <b>Intensive search</b><br>1000 x 300. Broad, slow search for demanding multilayer designs. |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|

PART II - HOW TO USE IT

# Follow progress, then trust the saved certified result

Press Start Campaign only after reviewing the current form. DOT snapshots those values, clears the previous plots, and uses the new targets immediately.



|                                                                                                                                                              |                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>During the run</b><br>Watch generation, ETA, live cross-section, every requested harmonic, every block margin, and the margin/residual/total-turn trends. | <b>At the end</b><br>A certified point must lie inside the acceptance box after dense re-evaluation. Select a point with reserve, not one exactly on every limit. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Saved artifact                              | Use it for                                                                   |
|---------------------------------------------|------------------------------------------------------------------------------|
| best_candidate_summary.png                  | One-page visual: cross-section, electromagnetic results and block geometry.  |
| best_candidate.json                         | Complete machine-readable certified representative.                          |
| best_candidate_geometry.csv                 | Per-block R, turns, phi and alpha for reconstruction.                        |
| final_pareto_frontier.png                   | Margin versus worst harmonic residual, colored by total turns.               |
| best_topology_designs/                      | Up to ten strong alternatives with different active-block topologies.        |
| pareto_candidates.json                      | Certified front plus near-feasible and search-front diagnostics.             |
| campaign.json + run_manifest.json + inputs/ | Exact settings, software versions, hashes and immutable conductor snapshots. |

**If nothing certifies**  
Do not immediately add generations. First inspect the near-feasible violation: geometry or current failures require a physically broader design space; a harmonic plateau may need more independent blocks or population; a small certification miss needs reserve.



PART III - RELIABILITY

# Measured agreement with ROXIE in 1,000 blind air-core cases

DOT 1.0.0 was compared through a licensed ROXIE service at localhost:8080: 500 two-layer and 500 four-layer layouts, all generated without reference block coordinates.

|                                                                                                 |                                                                                                |                                                                                                         |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Coverage</b><br>Aperture 20.004-39.968 mm; current 3.008-12.455 kA; 2-12 blocks; 5-44 turns. | <b>Conductors</b><br>Eight Nb-Ti, Nb3Sn and mixed layer patterns from the validated catalogue. | <b>Comparison</b><br>Identical R, turns, phi, alpha, current, conductor, symmetry and reference radius. |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

| Quantity         | Mean absolute | Median absolute | 95th percentile | Maximum absolute | Signed mean |
|------------------|---------------|-----------------|-----------------|------------------|-------------|
| Bore field       | 0.0197%       | 0.0156%         | 0.0525%         | 0.0988%          | +0.0181%    |
| Peak field       | 0.6709%       | 0.6727%         | 1.1866%         | 2.1149%          | +0.6235%    |
| Load-line margin | 0.1140 pp     | 0.1019 pp       | 0.2785 pp       | 0.6035 pp        | -0.0243 pp  |
| b3               | 1.2146 units  | 1.1119          | 2.7980          | 4.8190           | +0.8080     |
| b5               | 0.6507 units  | 0.5738          | 1.5252          | 2.4238           | -0.0132     |
| b7               | 0.2725 units  | 0.2314          | 0.7011          | 1.7254           | -0.0502     |
| b9               | 0.1035 units  | 0.0837          | 0.2610          | 0.5854           | -0.0033     |
| b11              | 0.0406 units  | 0.0351          | 0.1011          | 0.2547           | +0.0053     |

**Error definitions.** Field errors are relative to ROXIE. Margin error is an absolute percentage-point difference. Harmonic error is an absolute accelerator-unit difference because relative error is not meaningful near zero.

|                                                                                                                                                            |                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>What is strong</b><br>Bore-field parity is excellent. Peak field is close and DOT is conservative on average. Margin shows no material systematic bias. | <b>What needs reserve</b><br>b3 is the most sensitive cross-tool result: 2.80 units at the 95th percentile and 4.82 units maximum. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Practical reliability rule</b><br>Within the sampled 2D coil-only domain, DOT is a reliable pre-design engine. For a 5-unit field-quality requirement, prefer a DOT worst harmonic near 2 units or less. Independently verify candidates close to acceptance. The parity study does not validate iron, ends, magnetization, mechanics, protection, tolerances or out-of-domain layouts. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Evidence:** docs/validation/roxie\_parity\_1000\_results.csv and roxie\_parity\_1000\_summary.json. Deterministic seed 20260729; ROXIE runtime banner 23.6, git commit 26.1.0.b3.

## PART IV - DESIGN CHECKLIST

# A disciplined DOT campaign in ten checks

Use this page as the final design review before accepting a candidate for further engineering.

| #  | Check               | Question answered                                                                                  |
|----|---------------------|----------------------------------------------------------------------------------------------------|
| 1  | Scope               | The task is a 2D, straight, coil-only dipole pre-design.                                           |
| 2  | Cable data          | Every layer uses the intended supported conductor and temperature.                                 |
| 3  | Geometry convention | phi is zero at midplane; alpha is the absolute cable-frame angle.                                  |
| 4  | Design space        | Layer count, block/turn bounds and angle bounds do not exclude plausible solutions.                |
| 5  | Clearances          | Gaps and pole-turn proxy are physical values; numerical tolerance is only a classifier.            |
| 6  | Current             | Range or fixed-current mode is intentional and compatible with available turns.                    |
| 7  | Harmonics           | Reference radius and signed targets represent the intended air-core specification.                 |
| 8  | Search              | Quick is used for diagnosis; Design/Intensive and multiple seeds are used when warranted.          |
| 9  | Certification       | The chosen design is certified and has visible reserve in margin, harmonics, current and geometry. |
| 10 | Handoff             | Campaign JSON, cadata snapshots, manifest, candidate JSON and geometry CSV remain together.        |

## Ready-to-run example

**GUI:** choose campaign/dot\_cables.cadata for every layer and select a supported conductor.

**Command line:**

```
dot validate campaign/7T_NbTi_noiron_sample.json
dot optimize campaign/7T_NbTi_noiron_sample.json --output results/7T_sample
```

**Remember: a clean Pareto front is not a complete magnet. The next engineering steps must address iron, mechanical support and stress, protection, conductor degradation, tolerance robustness, end design, 3D effects and integration.**

## Primary references

K. Deb et al., "A fast and elitist multiobjective genetic algorithm: NSGA-II," IEEE Transactions on Evolutionary Computation 6(2), 2002, DOI 10.1109/4235.996017.

L. Bottura, "A practical fit for the critical surface of NbTi," IEEE Transactions on Applied Superconductivity 10(1), 2000.

DOT 1.1.0 source, tests, campaign archive and validation evidence dated 29 July 2026.